

VINITH BONILA

DATA & AI/ML · PYTHON DEVELOPER · PETROLEUM ENGINEER

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PROFESSIONAL SUMMARY

Python developer and Petroleum Engineering graduate who ships production - grade AI systems end to end. Deployed two live LLM applications backed by FastAPI services, Docker packaging, automated tests, and CI, plus upstream oil & gas domain knowledge from an ONGC field internship. Seeking early - career Data Analytics, AI/ML, or Python Developer roles.

TECHNICAL SKILLS

Programming: Python, SQL, C

AI/ML & NLP: RAG, LLMs, embeddings, BM25, cross - encoder reranking, spaCy, HuggingFace, sentence - transformers

ML & Evaluation: scikit - learn, TensorFlow / Keras, model evaluation (recall@k, MRR, faithfulness, F1)

Backend & DevOps: FastAPI, REST API design, OOP, unit testing, Docker, Git / GitHub, GitHub Actions CI, Streamlit

Data & BI: pandas, NumPy, EDA, SQL (MySQL / PostgreSQL), FAISS vector database, SQLite, Power BI, DAX, Tableau, Excel

Domain & Simulation: Upstream petroleum operations, SCADA / field data, reservoir simulation (CMG GEM, WinProp)

PROJECTS

- DocMind - AI Document Assistant (RAG)** | Python, FastAPI, Streamlit, FAISS, Groq LLM, Docker | [Live](#) 2026
- Built and deployed a production - grade RAG assistant answering questions over any uploaded PDF/DOCX/TXT with grounded, page - cited, streaming responses; served via FastAPI, containerized with Docker, and hardened with 24 automated tests and GitHub Actions CI.
 - Engineered hybrid dense (FAISS) and sparse (BM25) retrieval fused via Reciprocal Rank Fusion with cross - encoder reranking - recall@5 of 1.00 and MRR of 0.96; an evaluation harness measured faithfulness (0.78) and answer relevance (0.82) using LLM - as - judge.
- VINI AI - NLP - First Voice Assistant** | Python, Streamlit, spaCy, Whisper, SQLite, Groq API | [Live](#) 2026
- Built and deployed an NLP - first voice assistant resolving intent through semantic similarity - sentence - transformer embeddings with TF - IDF and fuzzy fallbacks - across a 19 - intent catalog at ~15 - 20 ms classification latency, replacing brittle keyword matching.
 - Engineered a modular NLU pipeline (spaCy NER with intent - aware slot extraction, dependency parsing, sentiment, language detection) with live web - search RAG on Groq Llama - 3.3 - 70B, a Whisper speech - to - text and TTS loop, SQLite persistence, and a 15+ skill handler router.
- Data Analytics - Zomato & Namma Yatri Insights** | Python, SQL, Power BI, Excel 2024
- Performed EDA with SQL and Python across multi - table ride and restaurant datasets, cleaning and joining raw records to surface demand, rating, and peak - period patterns.
 - Built interactive Power BI dashboards with DAX measures and drill - through filters, turning those patterns into an explorable view that supports data - driven decisions.
- Enhanced Oil Recovery & CO2 Sequestration Simulation** | CMG GEM, WinProp 2026
- Built a 3D compositional reservoir model to simulate long - term CO2 storage in deep saline aquifers, evaluating residual and solubility trapping mechanisms for safe geological storage (CCUS).
 - Ran compositional fluid modeling in WinProp on CO2 - brine interactions to characterize phase behaviour and solubility across reservoir pressure and temperature conditions.

EXPERIENCE

- Intern - Oil and Natural Gas Corporation (ONGC), Ankleshwar Asset** May - Jul 2025
- Analyzed upstream production facilities and SCADA - based real - time field - monitoring data systems across gas gathering, three - stage compression, dehydration, and metering, applying structured problem - solving across technical teams.

LEADERSHIP & ACHIEVEMENTS

- President - ShutrX, Photography Club 6.0, IIPE** (Jul 2025 - May 2026): led a 20+ member team across activities, event coverage, and social media.
- Public Relations Officer - SPE Student Chapter, IIPE** (Jul 2024 - Mar 2025): coordinated outreach, speaker communication, and event promotion.
- Cleared JEE Main & JEE Advanced** - secured admission to IIPE, an Institute of National Importance under MoPNG.

EDUCATION

- B.Tech, Petroleum Engineering - IIPE, Visakhapatnam** | CGPA: 6.29 2026
- Class XII (CBSE), PCM - Narayana College · 80.5% 2021

CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification - Stanford Online / DeepLearning.AI · Data Business Analytics Specialization - Jobaaj
- Excel Skills for Business - Macquarie University · SQL Gold Badge - HackerRank ·